



Imaging Spectroscopy Captures Climate-Dependent Phylogenetic Diversity Across Biomes

Presenter: Yu Wei

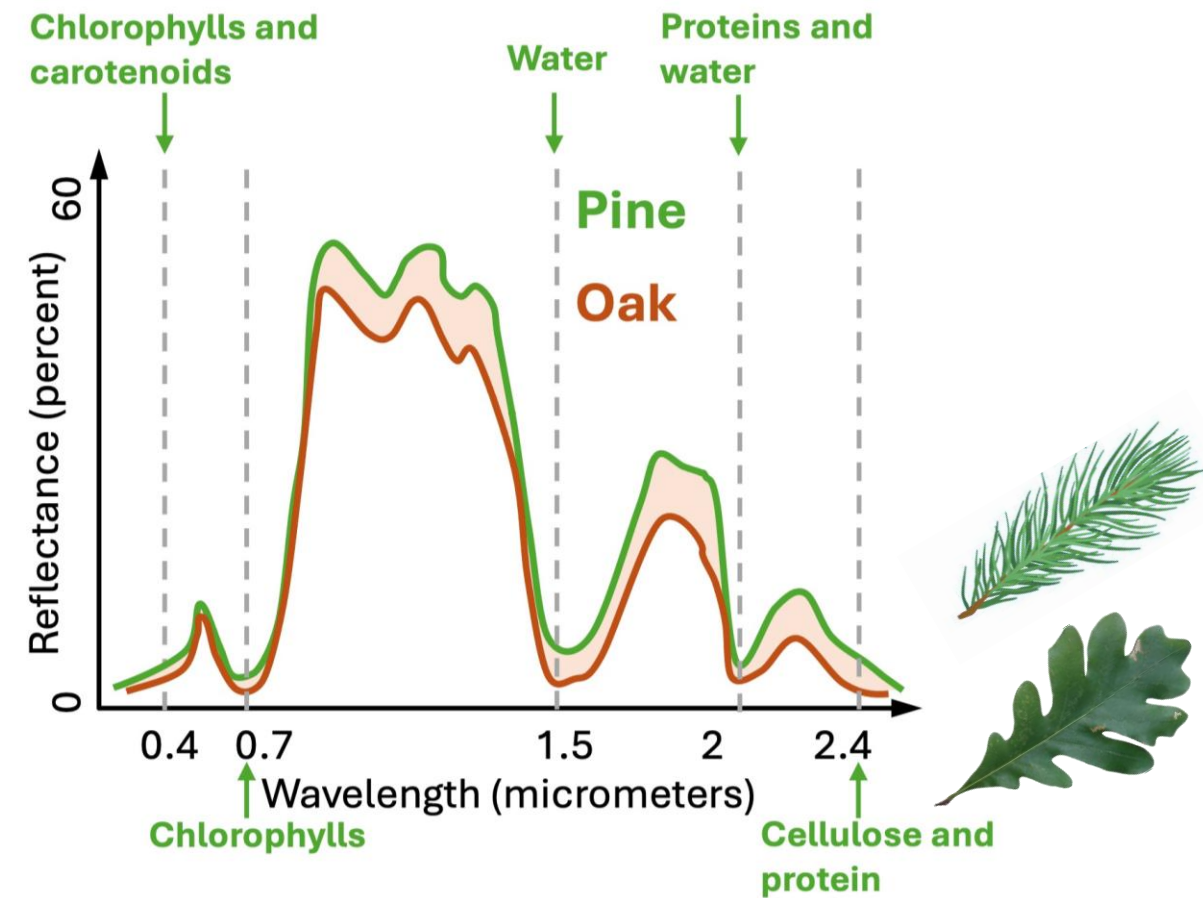
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of the Environment

*Yu et al., 2024 [Manuscript
available upon request]*

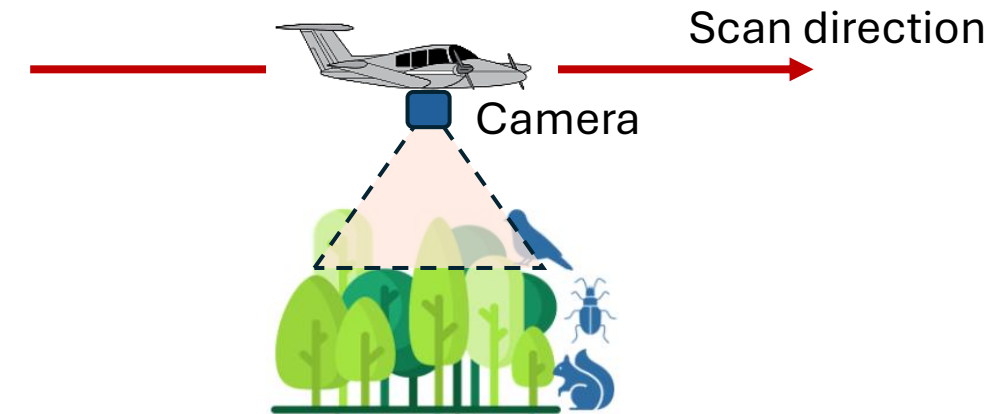
Imaging spectrometer

Plant leaf spectra

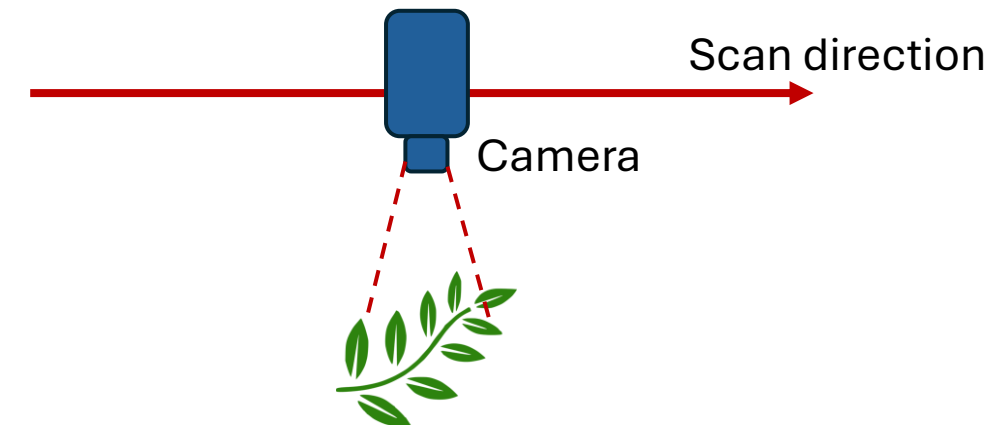


Imaging spectrometer

- Airborne/drone:



- Terrestrial spectrometer:



Plant Diversity Provides Essential Ecosystem Services

- Pollination
- Seed dispersal
- Invasion control



- Shelter
- Food supply
- Water
- Recreation
- ...

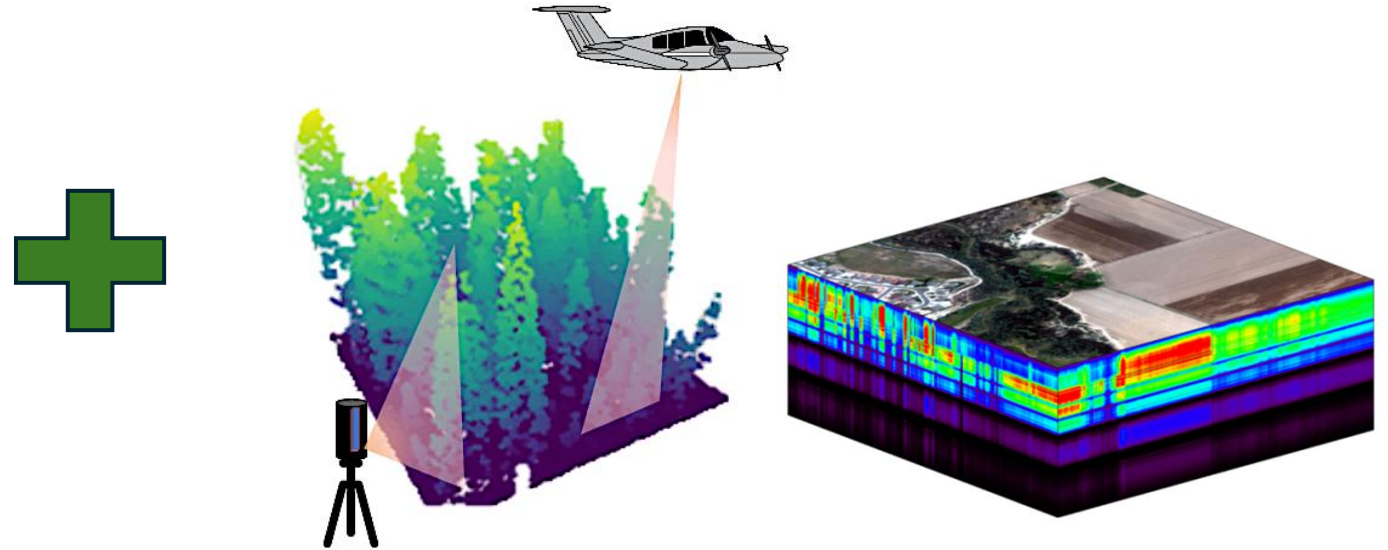


Integrate field sampling with spectroscopy

Sampling from the field



Spectral diversity from spectrometer



How can we link **plant diversity** with **spectral diversity**?

time-consuming
costly,
Limited spatial/temporal cover

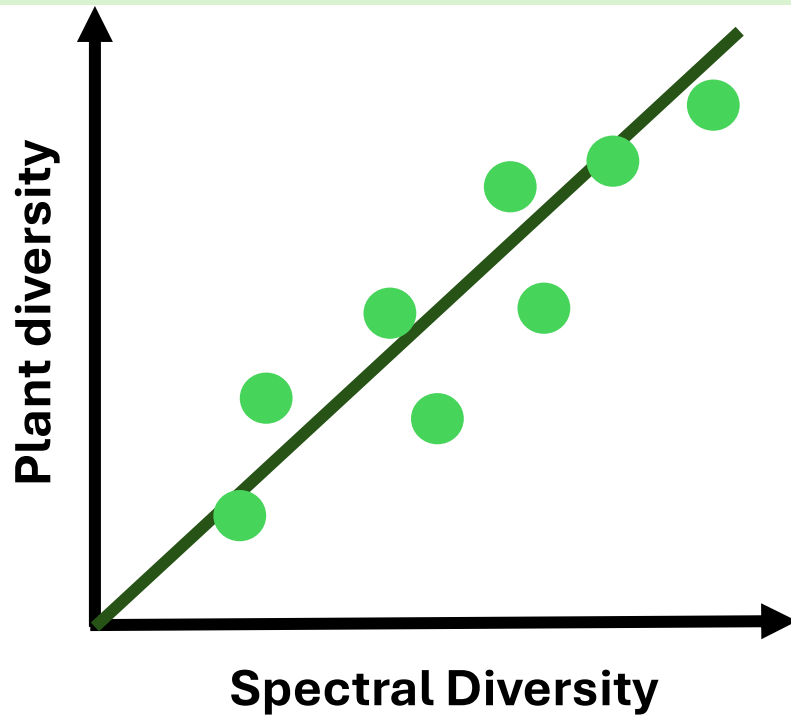
Large scale,
frequent,
wall-to wall coverage

Hypotheses

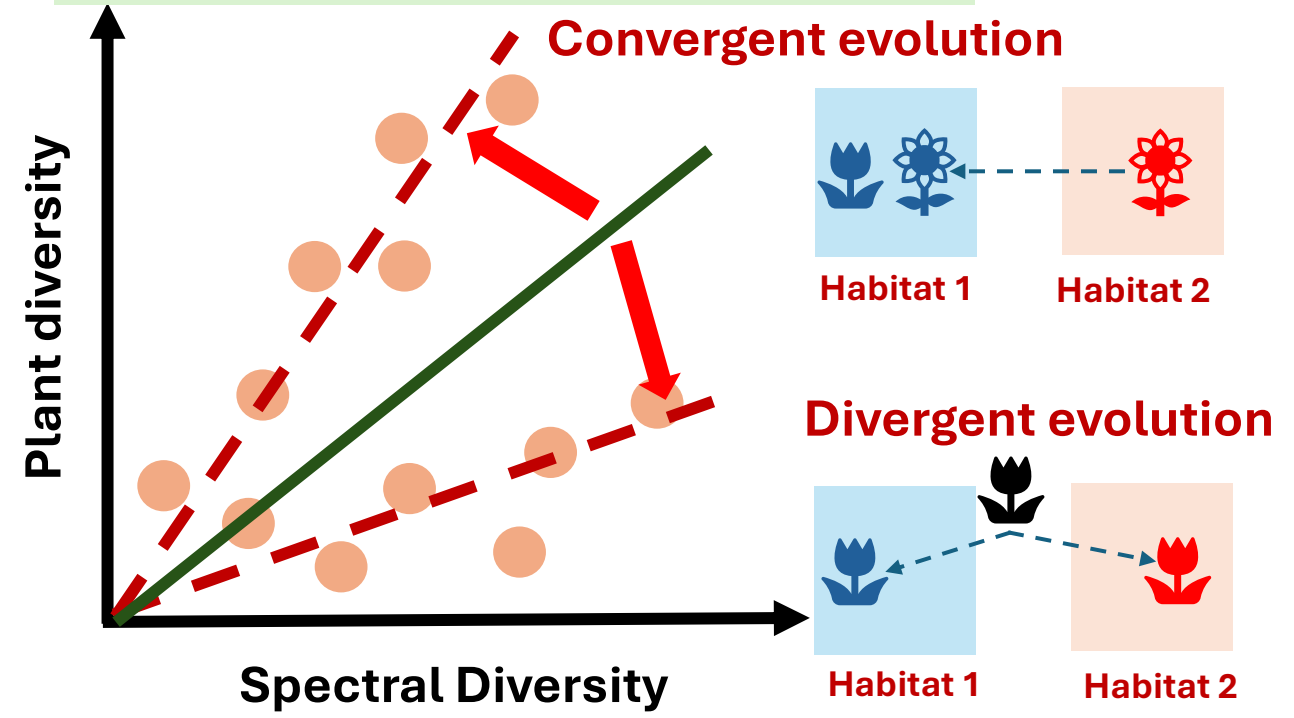
Spectral Variability Hypothesis (SVH): Spectral variation derived from remote sensing should positively correlate with **plant biodiversity**.

Hypotheses on the Spectral-Diversity-Biodiversity (SDB) relationship:

H1: strong & positive SDB relationships

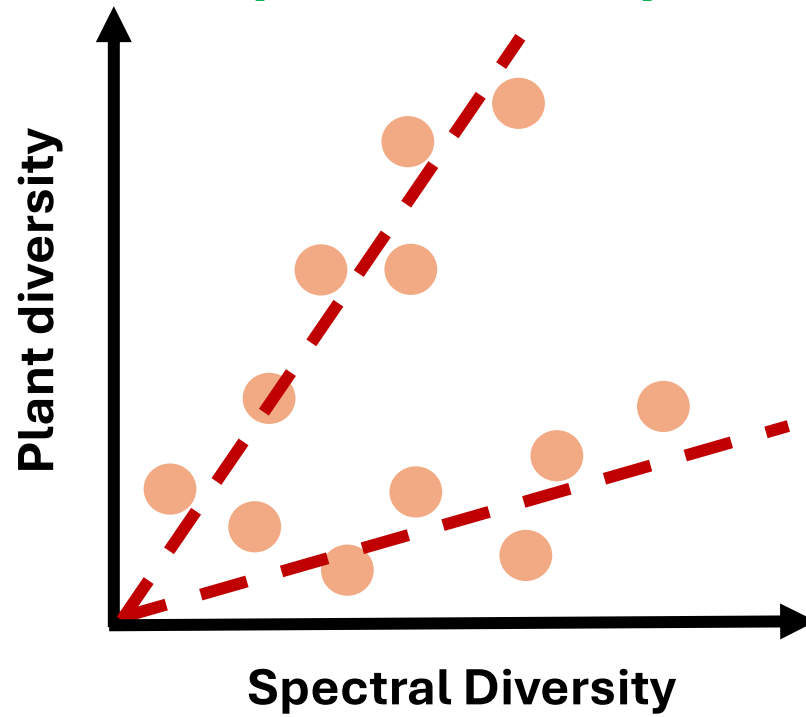


H2: weak / non-positive SDB relationships



Research questions

SDB relationships can be improved by controlling environmental factors:



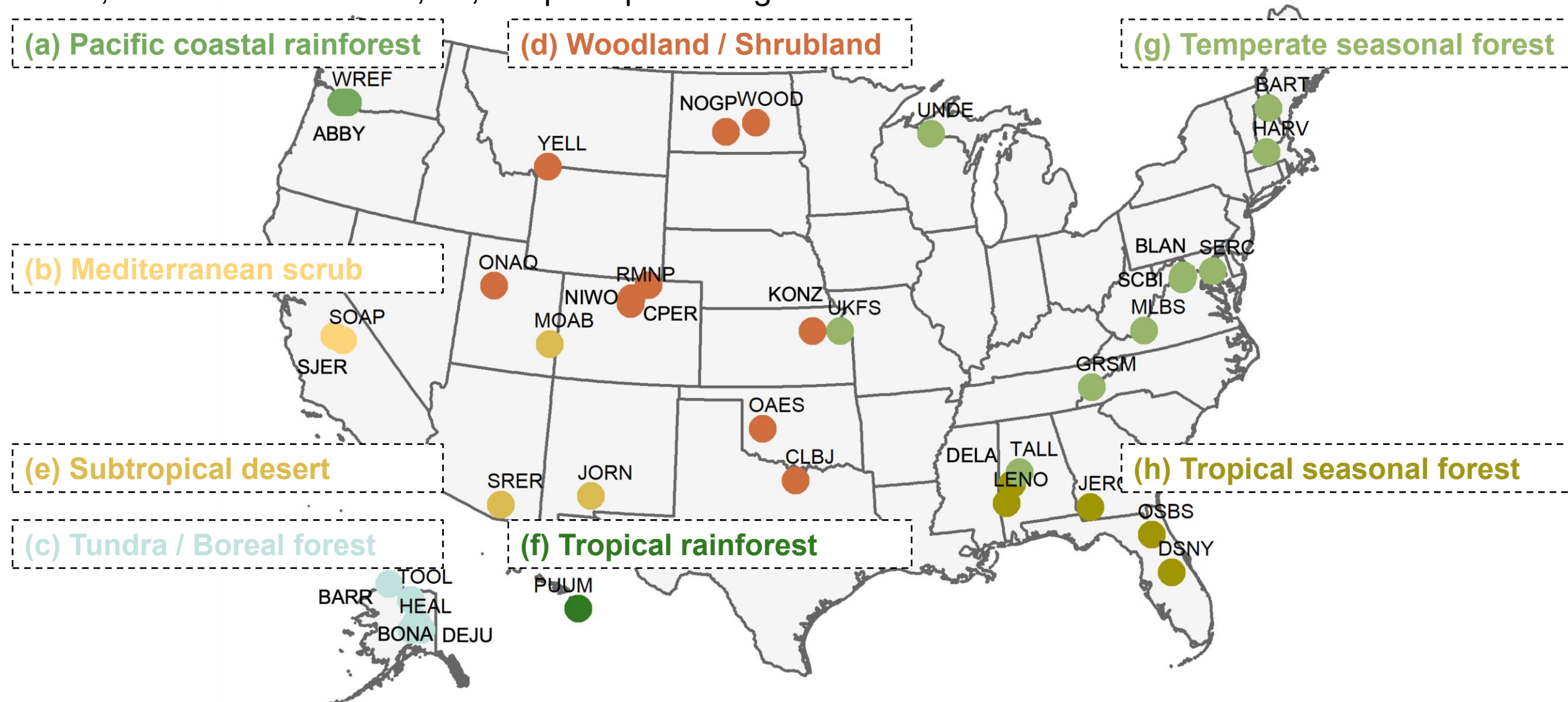
- Vegetation types 🌿 🍁
- temperature 🌡️
- Precipitation ☁️💧
- ...

Goal: use spectral diversity to predict plant diversity

- Which **dimensions of plant biodiversity** (e.g. taxonomic, phylogenetic, functional) can be more effectively predicted by the SDB relationship?
- How do **environmental factors** influence the SDB relationships across broad biogeographic regions?

Data and Materials

- 38 NEON terrestrial sites from 8 biomes
- field sampling data + hyperspectral imagery from the spectrometer
- 10,172 crown measures, 74,093 plant percentage cover records



1. SDB relationships varies across biomes

(a) Pacific coastal rainforest

Half negative

(b) Mediterranean scrub

All positive

(d) Woodland / Shrubland

Over 50% of the sites have positive relationships

(g) Temperate seasonal forest

Over 50% of the sites have positive relationships

(e) Subtropical desert

66% negative

(h) Tropical seasonal forest

Weak relationship

(c) Tundra / Boreal forest

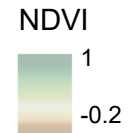
Weak relationship

(f) Tropical rainforest

All positive

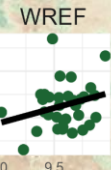
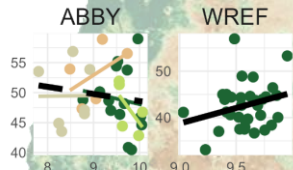
Plant diversity (PD)

Spectral diversity (Rao's Q)

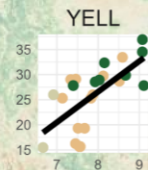


1. SDB relationships varies across biomes

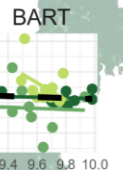
(a) Pacific coastal rainforest



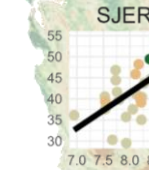
(d) Woodland / Shrubland



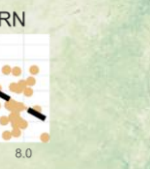
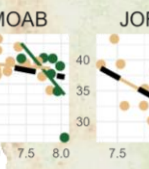
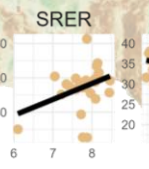
(g) Temperate seasonal forest



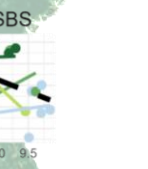
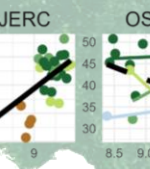
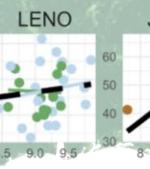
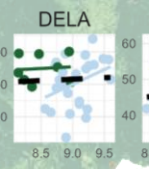
(b) Mediterranean scrub



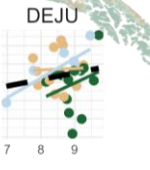
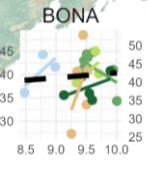
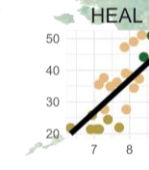
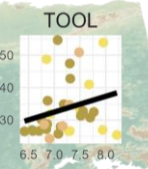
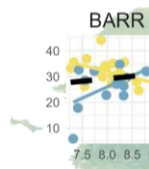
(e) Subtropical desert



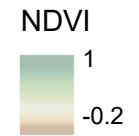
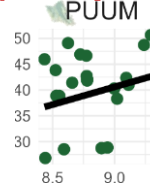
(h) Tropical seasonal forest



(c) Tundra / Boreal forest



(f) Tropical rainforest

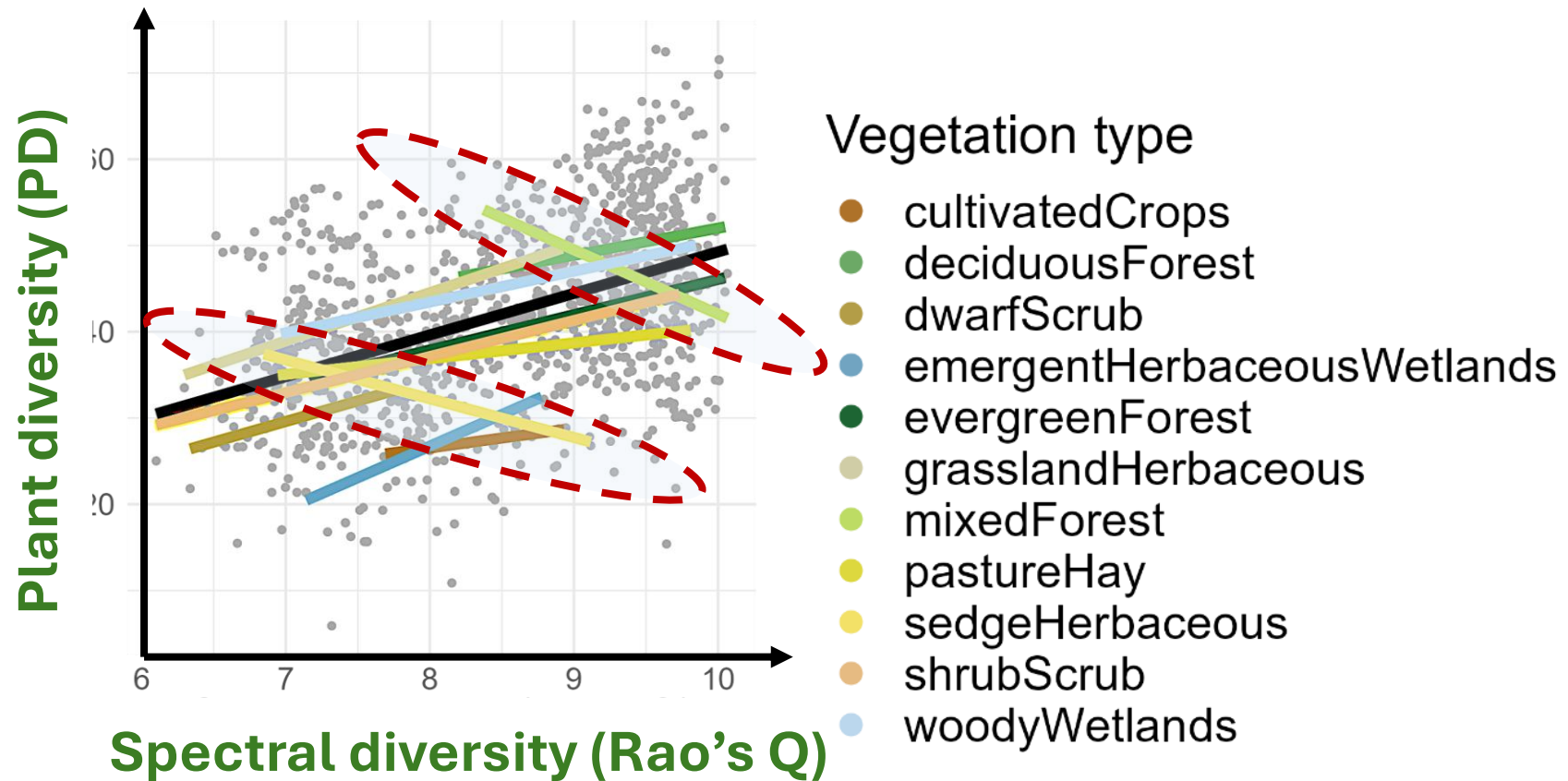


Plant diversity (PD)

Spectral diversity (Rao's Q)

2. SDB relationship varies across vegetation types

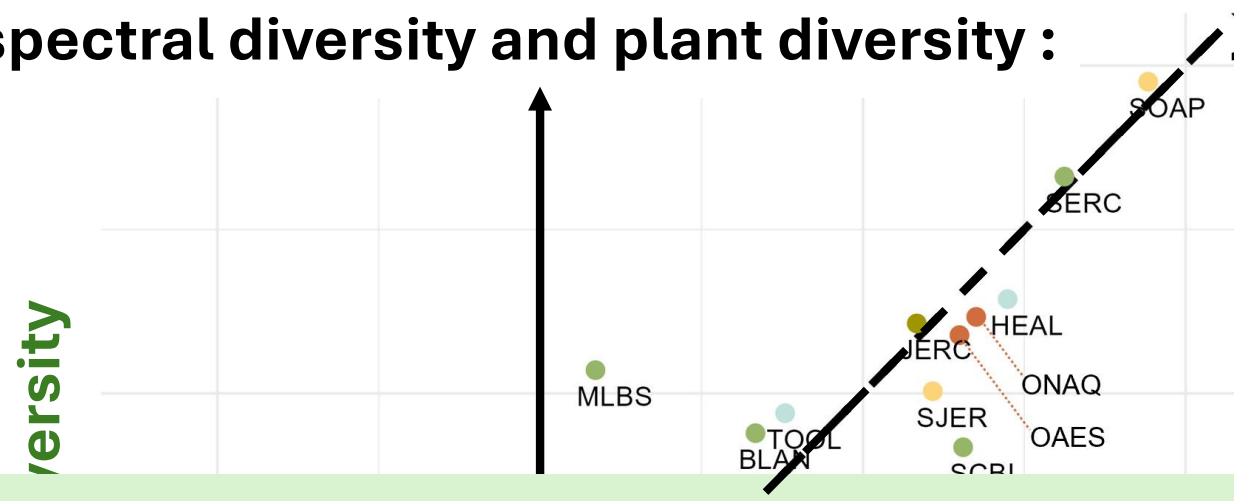
Overall relationships (of the 38 sites)



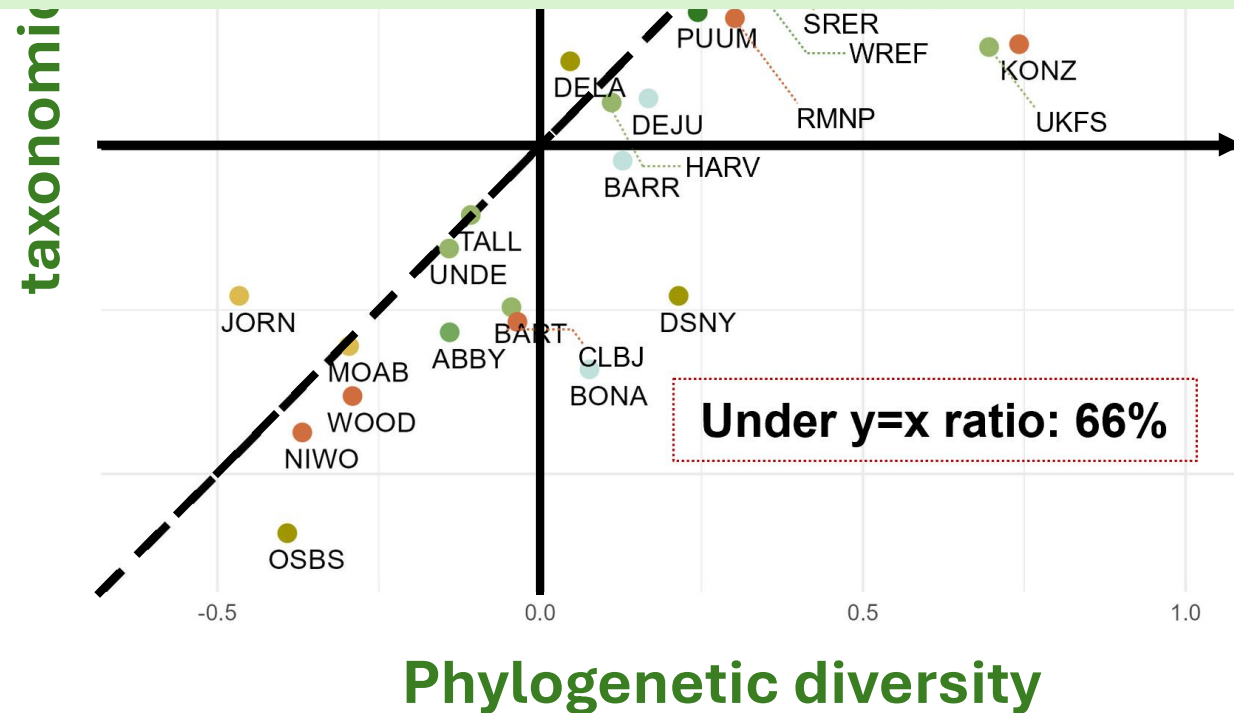
- Weak relationships: mixed forest, sedge & herbaceous

3. SDB relationships varies across of different dimensions of diversity

Correlation between spectral diversity and plant diversity :

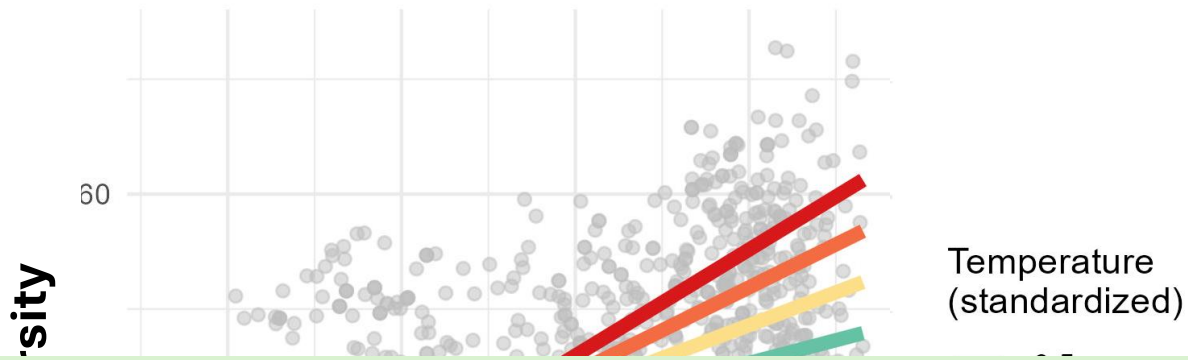


- Phylogenetic diversity is better captured than other dimensions

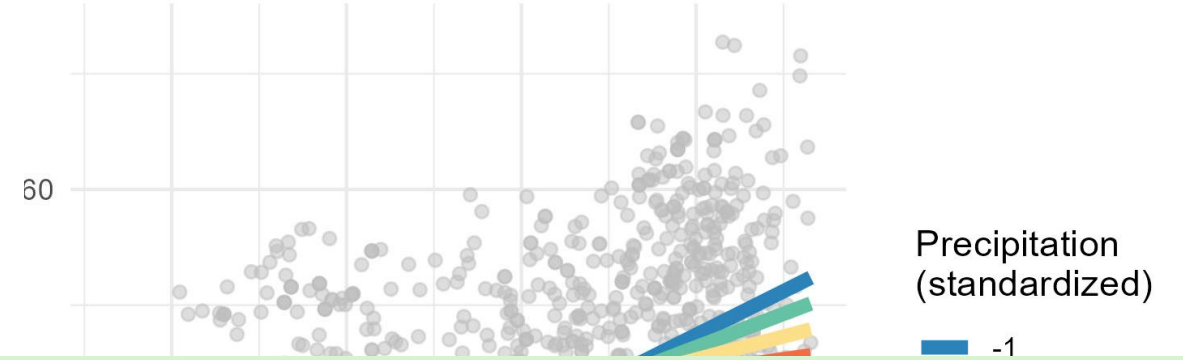


4. Climate factors modulate the SBD relationships

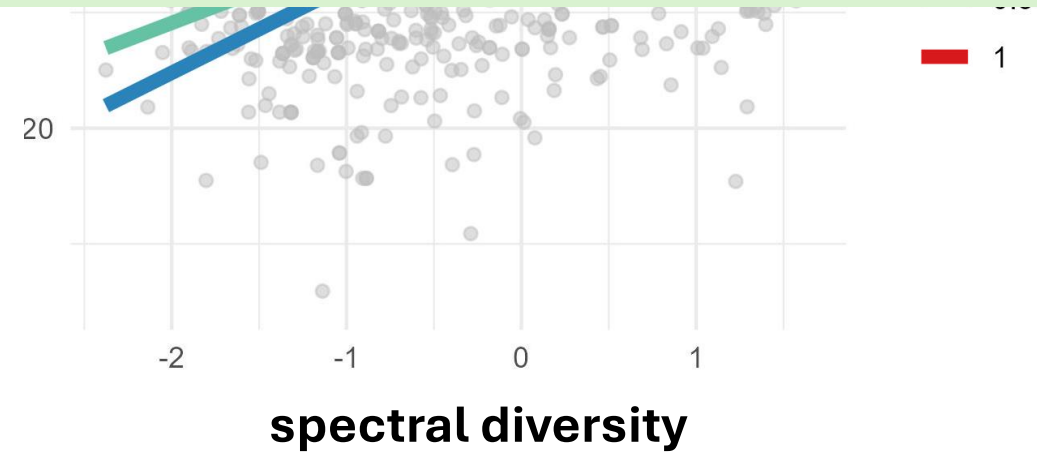
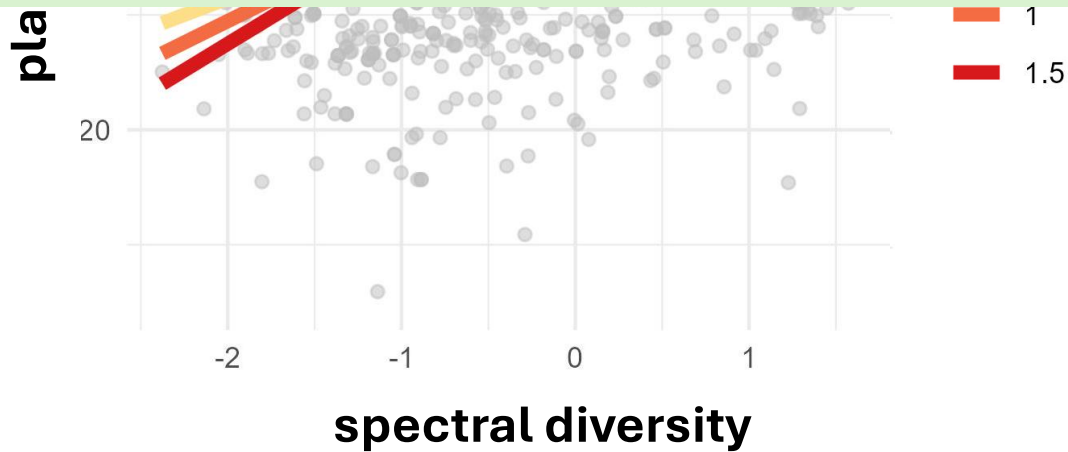
Control **precipitation**, and consider the influence of **temperature**:



Control **temperature**, and consider the influence of **precipitation**:



- Lower temperatures and higher precipitation can weaken the SDB relationships.



Conclusions

Recall our research questions:

1. Which dimensions of plant diversity can be better captured by spectral diversity?

- It better captures **phylogenetic diversity** compared to other diversity dimensions.

2. How do environmental factors modulate the SDB relationships across broad biogeographic gradients?

- **Mixed vegetation cover, sedge & herbaceous** could weaken the relationships.
- **Lower temperatures and higher precipitation** also weaken the relationships.

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